

Sample

Test window film

Objective of the Test

To clarify the performance of 3M Solar Control films 3M NV25 and No film by considering the glass with Temperature different by using temperature logger and BTU Meter.

Scope of the Test

1. Test Duration: This test was conducted from May21, 2019 to May24,2019.
2. Test Position: This test was conducted inside building as shown in Figure 1. Panes of glass windows were selected to conduct the experiment in order to compare the difference of heat transfer through glass windows between glass window with 3M NV25 and No film
3. Payback Period calculation by using eFilm software to simulate payback period.

Product specification: 3M NV25

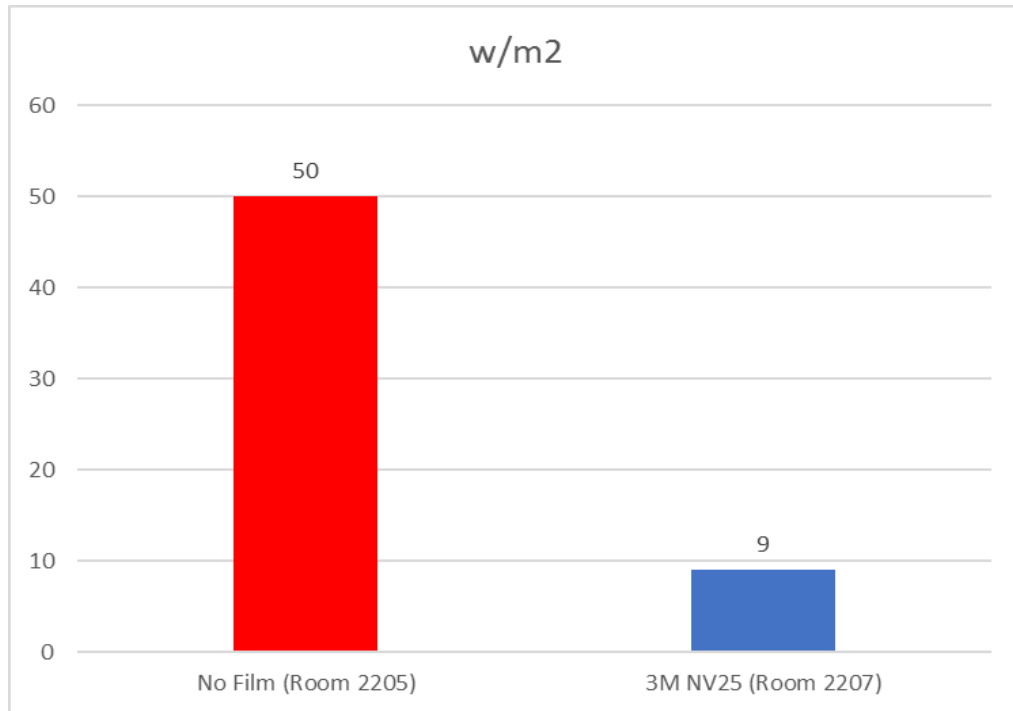
Product Benefits

- Warm tones complement natural light and defend against solar rays.
- A warm natural hue invites warmth and beauty to any room.
- Low interior and exterior reflectivity, especially at night.
- Great heat rejection for dual pane windows.
- 3M technology enhances color stability.
- Designed to reduce air conditioning costs by helping reduce excessive solar heat especially in hot sunny climates.
- Helps prolong the life and vibrancy of fabric, furniture and carpets.
- Helps reduce the chance of injury from flying glass.

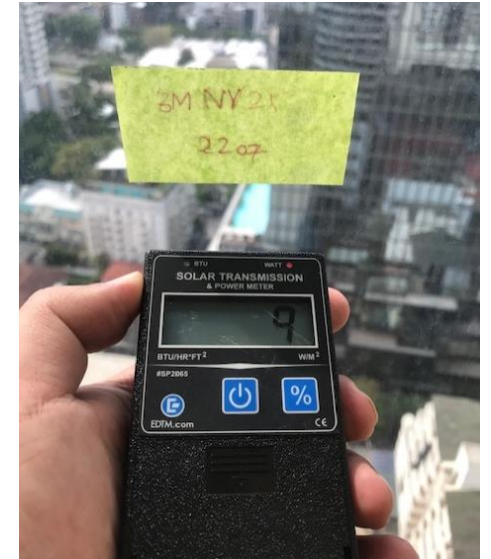
Product Performance & Technical Data

<u>NV25</u>								
	Single Pane		Tinted		Double Pane		Double tinted	
Film	1/4" Clear	NV25	1/4" tint	NV25	Dual 1/4" Clear	NV25	Dual 1/4" tint	NV25
Solar Heat Gain Coefficient	0.82	0.39	0.63	0.36	0.70	0.49	0.51	0.37
Visible Light Transmitted	89%	24%	53%	14%	79%	22%	47%	13%
Visible Light Reflected Interior	9%	7%	6%	7%	15%	8%	13%	8%
Visible Light Reflected Exterior	8%	19%	6%	10%	15%	24%	8%	12%
U Value	1.03	1.00	1.03	1.00	0.47	0.47	0.47	0.47
UV Block	38%	99%	NA	99%	NA	99%	NA	99%
Total Solar Energy Rejected	19%	61%	37%	64%	30%	51%	49%	63%
Glare Reduction	NA	73%	NA	74%	NA	72%	NA	73%
Heat Loss Reduction	NA	2%	NA	2%	NA	1%	NA	1%
Solar Heat Reduction	NA	52%	NA	43%	NA	29%	NA	27%

1) Solar Energy Measurement



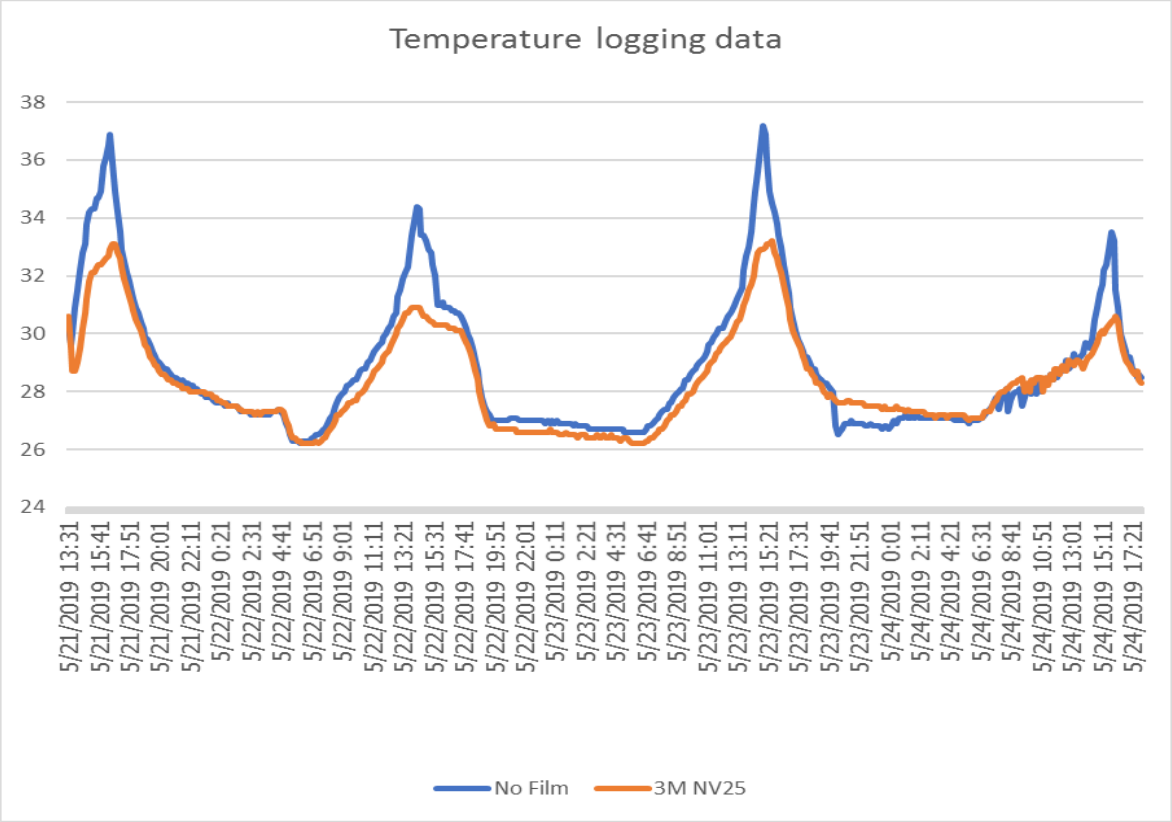
No Film



3M NV25

Solar Energy measured by Solar transmission meter

2)Record temperature



Date & Time	No Film	3M NV25	Diff(°C)
5/21/2019 16:31	36.9	32.9	4
5/22/2019 14:21	34.4	30.9	3.5
5/23/2019 15:01	37.2	32.9	4.3
5/24/2019 15:51	33.5	30.4	3.1

Maximum temperature reduced at peak temperature= 4.3 °C

3)Energy Simulation data by eFilm software

Building Details

- Total window area = 56,865 sq.ft.
- Electricity bill in 2018 = 41,615,608 THB
- Energy cost = 3.63 THB/kWh.



Window film cost for simulation

Window film cost for 3M NV25 = 120 THB/sq.ft.

Total cost = 6,823,800 THB.

Energy Simulation result by eFilm software

Targeting Variance for simulation =0.254%

☒ Targeting: Check the box if you would like to perform targeting

Utility Cost

	Target Utility Cost	Simulation Utility Cost
Electricity Utility Cost:	41,615,60 ₪	41,721,25 ₪
Fossil Fuel Utility Cost:	0 ₪	0 ₪

Iteration Summary

Target Variance: 10 %

Current Variance: 0.254 %

Maximum # of Iterations: 10

Current Iteration: 2

Ranges

Targeting Variables	Min	Max	
Internal Lighting Load	0	300	W/m?
Equipment Load	0	1,000	W/m?
Occupancy Load	1	1,000	m?/person
Core Infiltration	0	9.84	m?/h/m?
Perimeter Infiltration	0	39.37	m?/h/m?

Contributions and Targeting

☒ Interior Lighting Load ☐ Equal Weighting

☒ Equipment Load

☒ Occupancy Load

☒ Infiltration

Total: 100 %

Note: Sum of all weighting factors has to be equal to 100%

Targeting Details

Energy Simulation result by eFilm software

Building Energy Analysis Report

6/20/2019

Report generated by Efilm, an EnergyPlus based program.

Efilm version: v2.1.01

Simulation performed on: Monday, June 17, 2019

EnergyPlus version: 8.0

Project Information

Title: W Hotel
Address: Rama VI Road
City/State/Zip: Bangkok,

Building Owner

Name: Siriporn Chuchuen
Phone:
Address:
City/State/Zip: Bangkok

Analyst

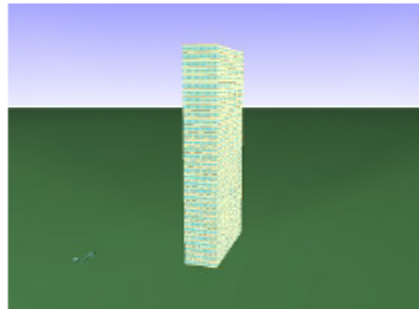
Name: Pichayapong Boonlue
Company: 3M
Address: Bangkok
City/State/Zip: Thailand

Architect

Name:
Phone:
Address:
City/State/Zip:

Building Information

Building Type: Hotel
Building Shape: Rectangular
Orientation: North
Number of Floors: 32
Floor-to-Floor Height: 3.85 m
Conditioned Area: 42,240 m²
Vintage: Post-1992
Category: Non-Residential



Dimensions

W1: 20 m L1: 66 m



Results Summary

Option	Gas	Electricity	Total (Site)	Total (Source)	Savings
Energy Totals					
Base Case	0 GJ	11,492,550 kWh	41,373 GJ	130,863 GJ	
3M NV25	0 GJ	11,167,964 kWh	40,205 GJ	127,167 GJ	3,696 GJ
Energy/Area					
Base Case	0 MJ/m ²	272.077 kWh/m ²	979.48 MJ/m ²	3,098.09 MJ/m ²	
3M NV25	0 MJ/m ²	264.393 kWh/m ²	951.82 MJ/m ²	3,010.59 MJ/m ²	87.5 MJ/m ²
Peak Demand					
Base Case	0 kW	1,975 kW			
3M NV25	0 kW	1,948 kW			
Energy Cost					
Base Case	0	41,721,299	41,721,299		
3M NV25	0	40,542,952	40,542,952		1,178,347
Energy and HVAC Equipment Cost					
Base Case	1,776	41,823,859	41,825,635		
3M NV25	1,765	40,635,254	40,637,019		1,188,615

Simple Payback

Option	Project Cost	Simple Payback Period
Base Case	0	
3M NV25	6,823,800	5.791 years

Payback period

Energy Simulation result by eFilm software

Glass area of each side of building

← Previous Next → Apply

☐ Daylighting Calculations ☒ Same Type For All Orientations ☐ Same Dimensions For All Orientations Reset

Fit Windows Using: **Total Windows Area**

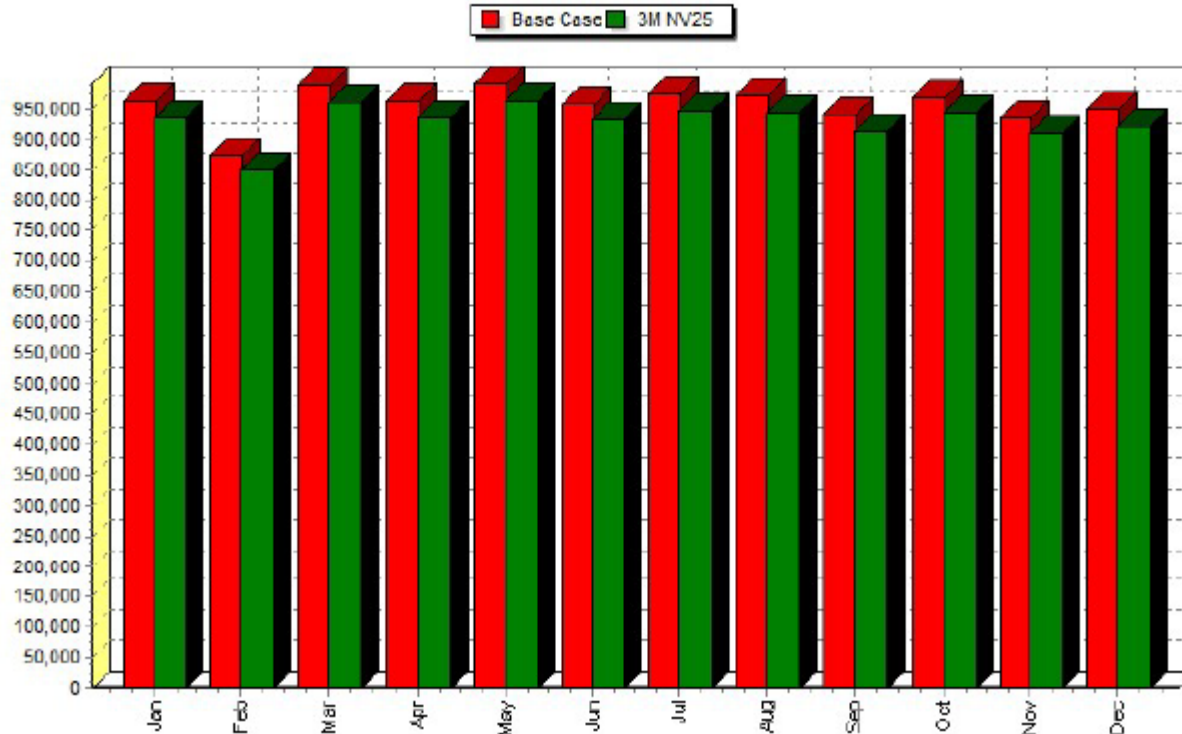
Orientation	Window Type	Width m	Height m	Total # / facade	Total Window Area / facade m²	External Shading
North	Dual Clear	1.21	2.284	415	1,147	None
East	Dual Clear	1.2	1.5	994	1,790.14	None
South	Dual Clear	1.21	2.032	415	1,020.35	None
West	Dual Clear	1.201	1.501	736	1,326.13	None
Building Totals:				2560	5,283.62	

Skylights

Window Type	Width m	Length m	Total #	Total Window Area m²
Skylights			0	0

Energy Simulation result by eFilm software

Monthly Electric Use [kWh]



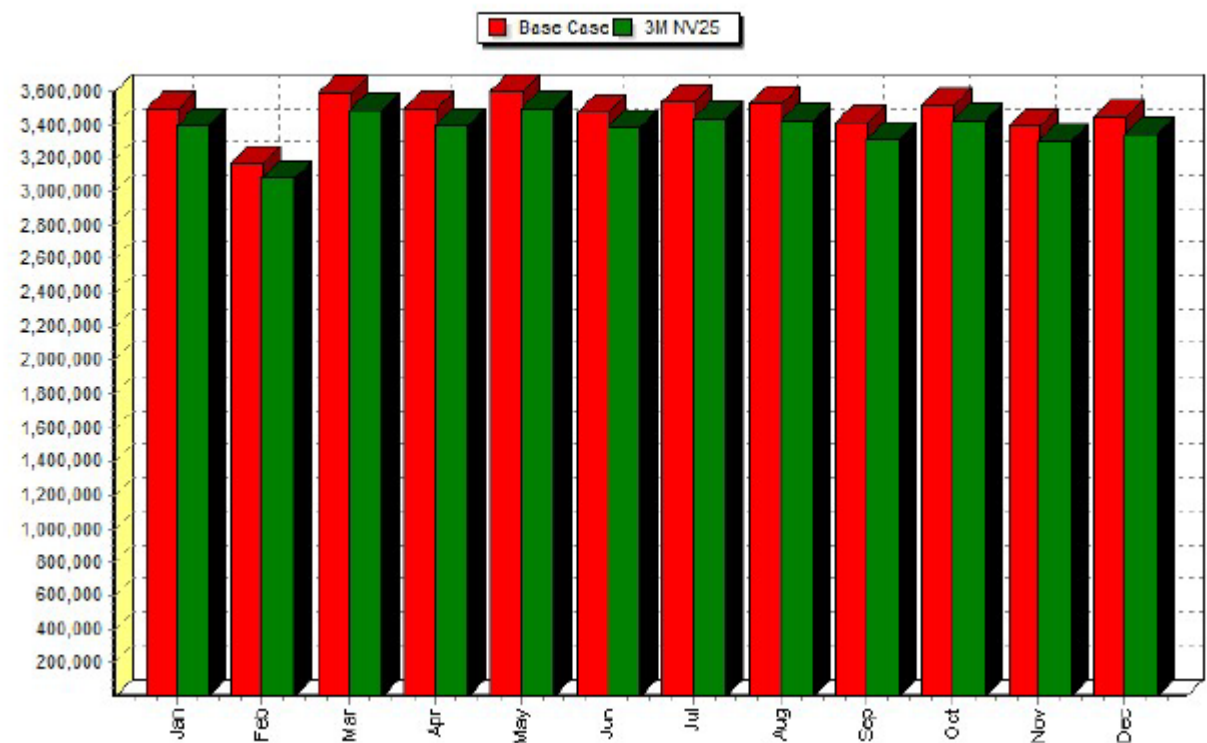
Yearly Electric unit saving

3M NV25 = 324,588 kWh

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Base Case	965,124	875,098	990,685	962,934	992,115	959,300	975,867	972,168	940,864	970,457	937,167	950,773	11,492,552
3M NV25	937,415	849,642	961,827	935,726	964,621	932,513	948,596	945,856	914,802	943,687	910,320	922,960	11,167,964
3M NV25 - Savings	27,709	25,456	28,858	27,208	27,494	26,788	27,270	26,312	26,063	26,770	26,847	27,813	324,588

Energy Simulation result by eFilm software

Monthly Electric Cost [฿]



Yearly Electric Cost saving

3M NV25 = 1,178,347 THB/Year

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Base Case	3,503,681	3,176,862	3,596,476	3,495,733	3,601,663	3,482,539	3,542,678	3,529,251	3,415,611	3,523,039	3,402,187	3,451,580	41,721,299
3M NV25	3,403,090	3,084,448	3,491,712	3,396,958	3,501,853	3,385,291	3,443,681	3,433,733	3,320,994	3,425,857	3,304,724	3,350,612	40,542,952
3M NV25 - Savings	100,591	92,414	104,764	98,775	99,810	97,248	98,998	95,519	94,617	97,182	97,463	100,968	1,178,347

Energy Simulation result by eFilm software

Payback Period

Simple Payback

Option	Project Cost	Simple Payback Period
Base Case	□0	
3M NV25	□6,823,800	5.791 years

Payback Period for 3M NV25 = 5.79 years.

Conclusion

1)The results of the solar energy measured by Solar Transmission meter

The Solar energy different between **No film** and **3M NV25** = **41 W/m²**
(from **50 W/m²** to **9 W/m²**)

2)The maximum different temperature of peak temp is 4.3 Degree C on May23,2019

3) Payback period calculated by eFilm software

- 3M NV25 = 5.79 years.

Thank you